

Final Project

STA 790: Special Topics - Extra Linear Models

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Abstract—Principal component analysis (PCA) regression is commonly used in mortality studies by valuation teams (insurance companies usually have a pricing team for premium calculation and a valuation team for experience studies) in the insurance industry due to the inherent multicollinearity in actuarial data. However, the related method, partial least squares (PLS) regression, with a lack of advocacy in the sector, often hinders its adoption. This paper applies both PCA and PLS regression to compare their predictive performance and potential advantages, emphasizing the benefit of incorporating the dependent information y into the training process through PLS.

I. INTRODUCTION

This project reviews the basic idea and structure of PCA and PLS taught in STA790 with some personal intuition with the background context provided and how SVD is used to find component loadings. We apply both methods to a mortality dataset from actuarial studies and compare their prediction performance. Since PCA only considers the variance in the predictors while ignoring their relationship with the response, it is expected that taking into account the covariance between $\mathbf{X}$ and $\mathbf{y}$, as PLS does, should lead to better predictive results. This project advocates for the PLS method in lieu of PCA, where the goal is accurate prediction of the mortality rate by considering $\arg \max \text{cov}(\mathbf{X}, \mathbf{y})$, rather than purely summarizing the variance of the predictors $\mathbf{X}$.

II. DATA

The dataset is originally from McDonald and Schwing (1973), “Instabilities of Regression Estimates Relating Air Pollution to Mortality,” *Technometrics*, vol. 15, pp. 463–482, which examines the relationship between air pollution and mortality rates using socioeconomic, weather, and pollution-related variables.

A. Variables Explanation and Intuition

- 1) **PRECReal** – Average annual precipitation (in inches): Regional rainfall; may influence pollution dispersion.
- 2) **JANTRReal** – Average January temperature (°F): Cold-season temperature; relevant for winter mortality patterns.
- 3) **JULTRReal** – Average July temperature (°F): Warm-season temperature; related to heat stress and ozone.
- 4) **OVR65Real** – Percent of population aged 65+ (1960): Indicator of elderly population vulnerability.
- 5) **POPNRReal** – Average household size: Reflects family structure and living density.
- 6) **EDUCReal** – Median years of education (age 22+): Socioeconomic indicator linked to health literacy.
- 7) **HOUSReal** – Percent of sound housing units: Housing quality measure; proxies for living conditions.
- 8) **DENSReal** – Population density (urban, 1960): Denser areas may face higher pollution exposure.
- 9) **NONWRReal** – Percent non-white population (urban, 1960): Demographic indicator relevant to exposure disparities.

- 10) **WWDKReal** – Percent in white-collar jobs: Socioeconomic status indicator.
- 11) **POORReal** – Percent of families with income < \$3000: Poverty indicator, linked to health risk.
- 12) **HCRReal** – Hydrocarbon pollution potential: Proxy for smog-forming emissions.
- 13) **NOXReal** – Nitric oxide pollution potential: Respiratory irritants and air quality metric.
- 14) **SO@Real** – Sulfur dioxide pollution potential: Industrial pollutant; affects lung health.
- 15) **HUMIDReal** – Avg. relative humidity at 1 PM (Affects pollutant behavior and air quality).
- 16) **MORTReal** – Age-adjusted mortality rate (per 100,000): Outcome variable; controls for population age structure.

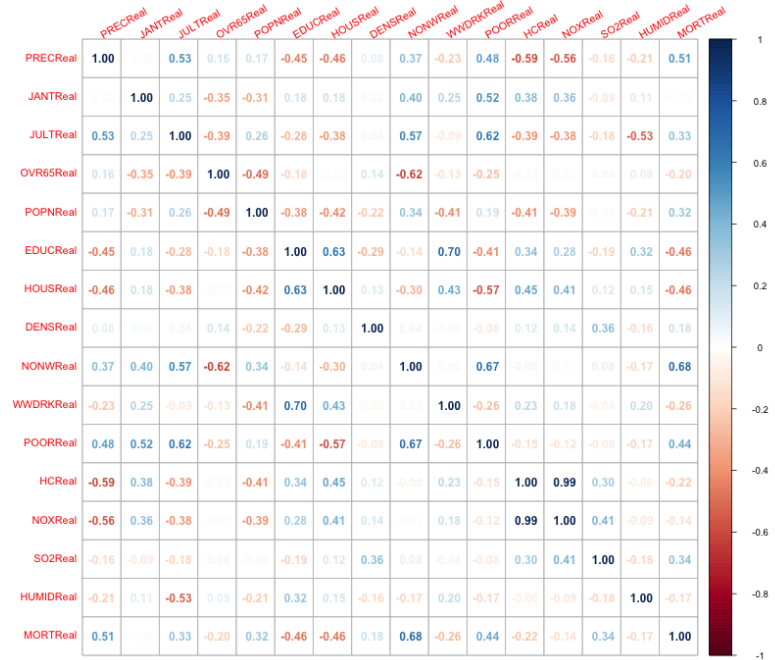


Fig. 1. Correlation Plot of Predictors

From Fig 1, we can observe that the highest correlation can reach 0.99, 0.7, which are high, and have high potential for multicollinearity. Hence, the following PCA and PLS dimension reduction methods should be the reasonable direction.

B. Training and Test Set

To train and compare the predictive performance of the PCA and PLS models, we randomly split the dataset into a 70% training set and a 30% test set with `ntest` and `itest` function:

```
ntest <- floor(n * 0.3)
itest <- sample(1:n, ntest)
```

The performance of each model is assessed using the prediction sum of squares (PSS), which measures the average squared difference between the observed and predicted values on the test set.

III. PCA MODEL

In our setting, $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote the standardized design matrix of $n = 42$ observations and $p = 15$ predictor variables, and let $\mathbf{y} \in \mathbb{R}^{n \times 1}$ denote the standardized response vector representing mortality rates. PCR regression first performs a low-rank approximation of the predictor space, we then fit a linear model based on how many components we want to keep.

A. Principal Component Decomposition

We can use the SVD method to find the PCs, which was mentioned during the class, on the normalized training design matrix $\mathbf{X}_{\text{train}}$:

$$\mathbf{X}_{\text{train}} = \mathbf{U}\mathbf{D}\mathbf{V}^\top,$$

where $\mathbf{U} \in \mathbb{R}^{n_{\text{train}} \times p}$ and $\mathbf{V} \in \mathbb{R}^{p \times p}$ are both orthogonal matrices, and $\mathbf{D} \in \mathbb{R}^{p \times p}$ is a diagonal matrix containing the singular values. The columns of $\mathbf{V}$ are the loading of PCs.

Let $\mathbf{V}_r \in \mathbb{R}^{p \times r}$ denote the matrix constructed by the first r columns of $\mathbf{V}$, which are the top r principal components from $\mathbf{V}$. These directions define the maximal-variance subspace (since we want to find the direction that maximizes the variance of $\mathbf{X}$) onto which the original predictors are projected. Hence, the final transformed training and test data are:

$$\mathbf{Z}_{\text{train}} = \mathbf{X}_{\text{train}}\mathbf{V}_r, \quad \mathbf{Z}_{\text{test}} = \mathbf{X}_{\text{test}}\mathbf{V}_r.$$

B. Model Fitting Process and Evaluations

As an OLS process is the rule of thumb for PCA regression, after we find the dimension reduction score, the model is then fit using the reduced training scores with all the same settings under regular OLS:

$$\mathbf{y}_{\text{train}} = \mathbf{Z}_{\text{train}}\boldsymbol{\beta}_r + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\beta}_r \in \mathbb{R}^r$ is the regression coefficient vector and $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ represents the residual errors. The model is estimated via ordinary least squares:

$$\hat{\boldsymbol{\beta}}_r = (\mathbf{Z}_{\text{train}}^\top \mathbf{Z}_{\text{train}})^{-1} \mathbf{Z}_{\text{train}}^\top \mathbf{y}_{\text{train}}.$$

Predictions on the test set are then obtained by:

$$\hat{\mathbf{y}}_{\text{test}} = \mathbf{Z}_{\text{test}} \hat{\boldsymbol{\beta}}_r.$$

Finally, the predictive performance of each reduced model is evaluated using the predictive sum of squares (PSS), which is the sum of squares of the difference between the predicted value and the true value:

$$\text{PSS}_r = \frac{1}{n_{\text{test}}} \|\mathbf{y}_{\text{test}} - \hat{\mathbf{y}}_{\text{test}}\|^2.$$

C. Model Selection

We will evaluate all the possible number of components from $r \in \{1, \dots, p\}$. The optimal number of components is selected by minimizing the predictive sum of squares:

$$r^* = \arg \min_{r \in \{1, \dots, p\}} \text{PSS}_r.$$

The final loading matrix $\mathbf{V}_{r^*}$ provides a reduced-dimensional interpretation of the linear relations between the original predictors and the response. This enables a parsimonious (fewer predictors) but still informative model for explaining regional variation in mortality based on the available predictors.

D. Explained Variance and Scree Plot

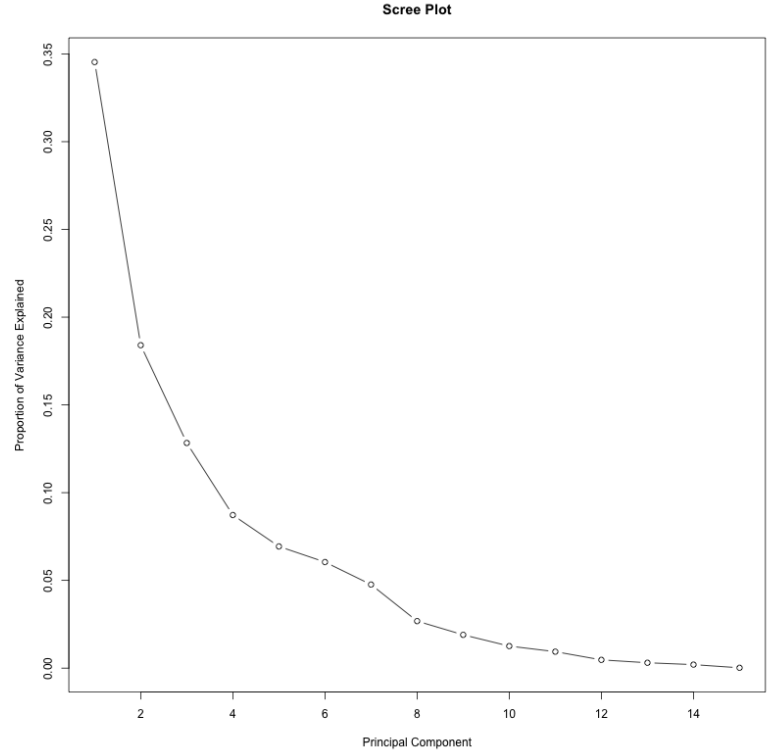


Fig. 2. Scree Plot

Fig. 2 shows that 8 components explain over 90% of the variance in $\mathbf{X}$, but in PCA regression, component selection should also consider predictive performance, not just explained variance.

E. Principal Component Loadings

The loading matrix $\mathbf{V}_r \in \mathbb{R}^{p \times r}$ indicates how each standardized predictor contributes (weights) to the derived principal components. The entry (j, k) represents the weight of the j -th predictor on the k -th principal component.

Table I presents the loadings for the top $r = 5$ components. Components with large-magnitude weights highlight variables that dominate that dimension. To be brief, we didn't show the full table, but replicating the code should provide you with the result: we can observe that the first principal component places high positive weight on `PRECReal` and `HOUSReal`, and strong negative weight on `HCREal` and `EDUCReal`. The intuition behind this scenario should be the contrast between resource access and healthcare burden in the insurance context.

	PC1	PC2	PC3	PC4	PC5
PRECReal	0.317	-0.043	-0.044	0.246	-0.063
JANTReal	-0.172	0.206	0.333	0.286	0.021
JULTRReal	0.283	0.192	0.266	0.017	-0.444
OVR65Real	-0.062	-0.193	-0.432	0.228	0.141
POPReal	0.345	0.154	0.064	-0.478	0.168
...	...	...	...	...	...
HUMIDReal	-0.083	-0.327	0.119	0.401	0.433

TABLE I

TOP 5 PCA PC LOADINGS FOR THE STANDARDIZED PREDICTORS.

IV. PLS MODEL

As was mentioned, instead of the maximization of the variance of $\mathbf{x}$, now, for each component $j = 1, \dots, r$, PLS constructs a weight vector $\mathbf{w}_j \in \mathbb{R}^p, p = 15$ by solving:

$$\mathbf{w}_j = \arg \max_{\|\mathbf{w}\|=1} (\mathbf{w}^\top \mathbf{X}^\top \mathbf{y})^2,$$

which maximizes the squared covariance between the score $\mathbf{t}_j = \mathbf{X}\mathbf{w}_j$ and the response vector $\mathbf{y}$. The corresponding loading vector $\mathbf{p}_j$ and regression coefficient c_j are computed as:

$$\mathbf{p}_j = \frac{\mathbf{X}^\top \mathbf{t}_j}{\mathbf{t}_j^\top \mathbf{t}_j}, \quad c_j = \frac{\mathbf{t}_j^\top \mathbf{y}}{\mathbf{t}_j^\top \mathbf{t}_j}.$$

After each component is extracted, both $\mathbf{X}$ and $\mathbf{y}$ are deflated(the less information could be extracted):

$$\mathbf{X} \leftarrow \mathbf{X} - \mathbf{t}_j \mathbf{p}_j^\top, \quad \mathbf{y} \leftarrow \mathbf{y} - c_j \mathbf{t}_j.$$

This highlights an important idea mentioned in STA790: subsequent components are orthogonal and capture a new covariance structure.

Once r components are obtained, the response is predicted using:

$$\hat{\mathbf{y}} = \mathbf{T}_r \boldsymbol{\beta}_r = \sum_{j=1}^r c_j \mathbf{t}_j,$$

where $\mathbf{T}_r = [\mathbf{t}_1, \dots, \mathbf{t}_r]$ and $\boldsymbol{\beta}_r = [c_1, \dots, c_r]^\top$. Then, again, we evaluate predictive performance by PSS:

$$\text{PSS}_r = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (y_i - \hat{y}_i)^2 = \frac{1}{n_{\text{test}}} \|\mathbf{y}_{\text{test}} - \hat{\mathbf{y}}_{\text{test}}\|^2,$$

and choose the value of r that minimizes PSS_r .

A. PLS Loadings for $r = 5$ Components

Table II presents the PLS loading matrix $\mathbf{P} \in \mathbb{R}^{p \times r}$ corresponding to the PLS model with $r = 5$ latent components. Very close to the explanation of the loadings for PCA, each column of the matrix represents a loading vector for a PLS component, showing the contribution(weight) of each original predictor to that component. We could explain that larger absolute values indicate predictors with a strong influence on the corresponding latent dimension.

V. COMPARISONS

To assess the predictive performance of PCR and PLS regression, we evaluate the PSS across different numbers of components. Fig. 3 shows that PLS regression consistently achieves lower prediction error compared to PCA regression, especially when the number of components is small. This reflects the nature of PLS, which incorporates response information(supervised) during component construction, allowing it to find directions in the predictor space that are more relevant to the response variable.

	Comp 1	Comp 2	Comp 3	Comp 4	Comp 5
PRECReal	0.327	-0.186	0.302	0.100	0.332
JANTReal	-0.093	0.310	-0.223	-0.131	-0.247
JULTRReal	0.331	-0.179	-0.237	0.297	-0.082
OVR65Real	-0.117	-0.036	0.186	-0.340	0.715
POPReal	0.392	-0.216	-0.079	0.361	-0.481
...	...	...	...	...	...
HUMIDReal	-0.156	0.042	0.625	-0.507	-0.213

TABLE II

TOP 5 PLS LOADING VECTORS FOR THE STANDARDIZED PREDICTORS. EACH COLUMN CORRESPONDS TO A LATENT COMPONENT.

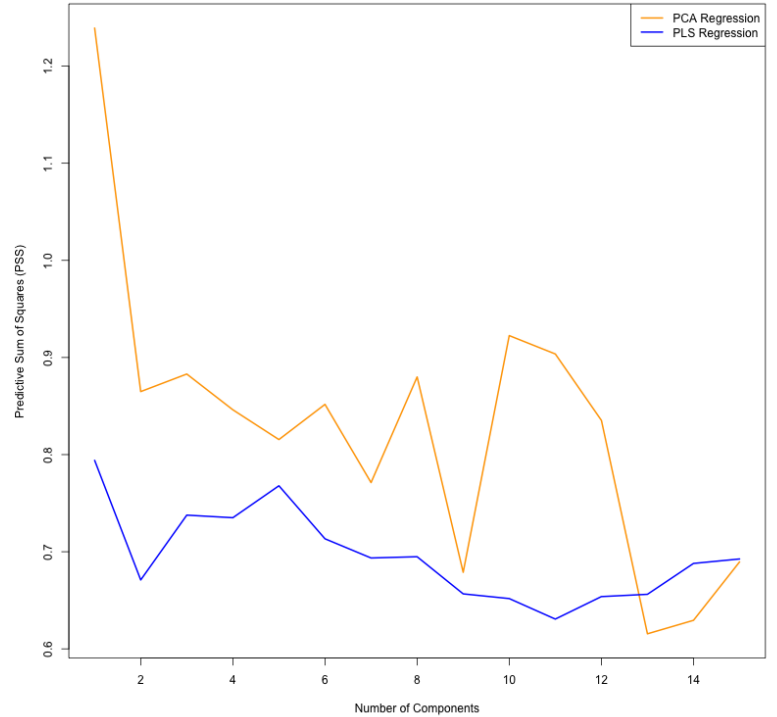


Fig. 3. PSS versus number of components for PCA and PLS regression. Lower PSS indicates better predictive performance.

VI. CONCLUSIONS

For this application to this actuarial data, it is apparent that PLS outweighs PCA, and this proves the words we discussed in STA790 that there is no reason to use PCA if PLS is possible. However, sometimes, what we want to do is more than the regression; what if we don't have the y variable, or what if we only want to use PCA to summarise some of the variables and only want to observe the characteristics of the specific loadings, then PCA might still be useful. Furthermore, with predictive performance as our target, I will still say that using the PLS model to predict the mortality rate for the insurance sector is still a very good one. In terms of PSS, PLS should be the right one to use when conducting the experience studies for all other actuarial predictions, not only the mortality rate.